

Assignment 3:

Calculate MAE and MSE errors for the given data.

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Table-

Year	Advertising (x)	Sales (y)	xy	x ²	y ²
1	23	651	14973	529	423801
2	26	762	19812	676	580644
3	30	856	25680	900	732736
4	34	1063	36142	1156	1129969
5	43	1190	51170	1849	1416100
6	48	1298	62304	2304	1684804
7	52	1421	73892	2704	2019241
8	57	1440	82080	3249	2073600
9	58	1518	88044	3364	2304324
Σ	371	10199	454097	16731	12365219

$$\therefore m = \frac{n \sum xy - \sum x \sum y}{n \cdot \sum x^2 - (\sum x)^2} = \frac{(9 \times 454097) - (371 \times 10199)}{(9 \times 16731) - (371)^2}$$
$$= 23.42$$

$$\therefore c = \frac{\sum y \cdot \sum x^2 - \sum x \cdot \sum xy}{n \cdot \sum x^2 - (\sum x)^2} = \frac{(10199 \times 16731) - (371 \times 454097)}{(9 \times 16731) - (371)^2}$$
$$= 167.68$$

Now,

x	Actual, (y)	Predicted, ($\hat{y}$)	Residuals	(Residuals) ²
23	651	706.34	-55.34	3062.52
26	762	776.6	-14.6	213.16
30	856	870.28	-14.28	203.92
34	1063	963.96	99.04	9808.92
43	1190	1174.74	15.26	232.87
48	1298	1291.84	6.16	37.95
52	1421	1385.52	35.48	1258.83
57	1440	1502.62	-62.62	3921.26
58	1518	1526.04	-8.04	64.64

$$\therefore \sum (\text{Residuals})^2 = 18804.07$$

$$\therefore \sum |\text{Residuals}| = 310.82$$

$$\begin{aligned} \therefore \text{Mean Absolute Error, MAE} &= \frac{1}{n} \sum_{i=1}^n |\text{Residuals}| \\ &= \frac{1}{9} \times (310.82) \\ &= 34.54 \\ &\quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} \therefore \text{Mean Squared Error, MSE} &= \frac{1}{n} \sum_{i=1}^n (\text{Residuals})^2 \\ &= \frac{1}{9} \times (18804.07) \\ &= 2089.34 \\ &\quad (\text{Ans}) \end{aligned}$$